

BMJ Open How do doctors build knowledge around obstetric emergencies? A scoping review

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ABSTRACT

Background Obstetric emergencies (OEs) are high-stakes events but, despite their importance, how doctors build knowledge relating to them remains poorly understood.

Objectives Explore the international literature on knowledge-building around OEs with the goal of informing future UK training and workforce strategies.

Search strategy Subject headings, free-text terms and synonyms relating to OEs were included. Searches were conducted in April 2024 and updated in June 2026 across MEDLINE, Embase, PsycINFO (all via Ovid), ERIC (via EBSCOhost) and Scopus.

Selection criteria English language studies, published after 2016, addressing OE knowledge-building in obstetrics and gynaecology (O&G) doctors across all grades, were included.

Data collection and analysis Screening was undertaken in two phases (title and abstract and full text) with reasons for exclusion documented. A 10% sample was double-screened and discrepancies resolved through discussion. A structured data extraction tool was refined before full extraction. Thematic synthesis was conducted inductively.

Main result Findings from 369 papers were synthesised. Knowledge-building sources clustered into training, context and experience, with formal training remaining the primary focus of the literature. Barriers to learning were both structural and cultural. Inequalities permeated all domains, driving inequities in outcomes.

Conclusion O&G doctors can draw on multiple sources to build OE knowledge, yet workplace-based learning (WPBL) remains a significant gap. Separating learning from clinical practice wastes knowledge-building opportunities inherent in everyday care. WPBL approaches within OE must be developed and rigorously evaluated to realise their full potential.

INTRODUCTION

Obstetric emergencies (OEs) are high-stakes, time-critical events that arise during pregnancy, birth or the immediate postnatal period and demand coordinated, skilled responses to prevent severe maternal and neonatal harm. For obstetrics and gynaecology (O&G) doctors, developing and applying knowledge in these situations is essential for clinical safety, effective teamwork and patient trust. Yet, the processes through

STRENGTHS AND LIMITATIONS OF THIS STUDY

- ⇒ An exhaustive search strategy and rigorous scoping review methodology were applied, with meaningful patient and public involvement throughout, strengthening the relevance and interpretive depth of the synthesis.
- ⇒ Screening was undertaken primarily by a single reviewer, with a random 10% sample independently double-screened; this was a pragmatic approach, adopted because of the large volume of eligible studies, but may have introduced a small risk of selection bias.
- ⇒ Grey literature and citation searching were not undertaken due to the large volume of eligible studies, which may have excluded relevant perspectives—particularly contextual or experiential knowledge that may sit outside formal academic publications.
- ⇒ The review focused specifically on obstetrics and gynaecology (O&G) doctors rather than fully interdisciplinary teams, which may limit insights into collaborative practice in obstetric emergencies.
- ⇒ The wide range of included studies enabled identification of transferable lessons across international contexts; however, the breadth of the review may have limited deeper exploration of local, context-specific implementation challenges.

which O&G doctors build knowledge remain poorly understood. In this review, we conceptualise *knowledge-building* as the process through which O&G doctors acquire, develop and apply understanding relevant to OEs. This includes formal education and training, learning from clinical experiences, engagement with evidence and guidelines and insights gained through working with colleagues and patients. We intentionally adopt a broad, constructivist perspective because, while individual learning within OEs is important, we recognise that knowledge in OE is shaped by workplace conditions, available resources and professional interactions. Rather than conceptualising knowledge as something transmitted through formal education alone, constructivist



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learning theory¹ views knowledge as continually developed through interaction between individuals, clinical practice and organisational context. Within this review, competence and expertise are understood as outcomes that may emerge from these processes, whereas our focus is on how knowledge is constructed, refined, shared and applied throughout clinical practice. This perspective, therefore, allows us to examine the multiple influences on knowledge-building across training, workplace and clinical contexts.

The *Better Births* report² marked a major policy shift in UK maternity care, calling for safer systems through improved staffing, a stronger safety culture and inter-professional training—all measures with clear relevance to OEs, where rapid escalation, coordinated teamwork and reflective learning are essential. These reforms should have strengthened emergency preparedness and improved outcomes; however, since its publication, maternal mortality has risen, particularly among women from marginalised and vulnerable groups,³ highlighting persistent gaps between policy intent and clinical reality. Subsequent reviews, including the Ockenden and Kirkup reports,^{4 5} have documented systemic failures in NHS maternity services, including insufficient emergency preparedness, poor communication and hierarchical team dynamics, all of which have significant implications for how O&G doctors are supported to respond to, and learn from, OEs. Although this review draws on international literature, it is undertaken to inform UK maternity education, training and workforce strategies in response to these ongoing challenges.

Knowledge-building in OEs is likely to be complex and context-dependent, but no review has systematically mapped the literature to understand what sources O&G doctors can draw on to build knowledge around OEs, nor what barriers they face in doing so.

Objective

To explore the international literature on how O&G doctors build knowledge relating to OEs, with the goal of informing future UK training and workforce strategies.

RESEARCH QUESTIONS

1. What sources can O&G doctors use to build knowledge relating to OEs?
2. What are the barriers and enablers to O&G doctors building or applying this knowledge?

METHODS

This scoping review was conducted in accordance with the methodological framework developed by Arksey and O'Malley.⁶ The review was prospectively registered with the Open Science Framework (https://osf.io/dtfp9/overview?view_only=a33cf24f4b824738b456b7cea25f90b8) and is reported in accordance with the Preferred Reporting Items for Systematic Review and Meta-Analysis

extension for scoping reviews (PRISMA-ScR) checklist (online supplemental file S1). This methodology allows a comprehensive and flexible approach to mapping the diverse literature relating to how O&G doctors acquire and apply knowledge around OEs; capturing heterogeneity in study types and perspectives and identifying research gaps.

Search strategy

The search strategy was developed using the PCC (Population–Concept–Context) framework and informed by team expertise. It was iteratively refined with Patient and Public Involvement input and tailored to each database with the assistance of an academic medical librarian. Subject headings, free-text terms and synonyms relating to OEs and O&G doctors were included. Searches were conducted in April 2024 and updated in June 2026 across MEDLINE, Embase, PsycINFO (all via Ovid), ERIC (via EBSCOhost) and Scopus. Online supplemental file S2 includes the full search strategy.

Eligibility criteria

Studies were included if they addressed knowledge-building in OEs by O&G doctors, across all grades. Inclusion was limited to English-language and literature published from 2016 onwards, in line with the policy shift instigated by *Better Births*.

Screening and data extraction

Screening was undertaken in *Rayyan*, a web-based systematic review screening platform, initially by one reviewer (HS) in two phases (title/abstract screening followed by full-text screening) with reason for exclusion documented.⁷ A random 10% sample was independently double-screened (LM), and discrepancies resolved through team discussion. A structured Microsoft Excel data extraction tool was piloted and refined before full extraction. Data were coded inductively by one reviewer (HS) using Microsoft Excel in accordance with the iterative approach described by Arksey and O'Malley.⁶ Codes were developed from the extracted data and progressively refined into broader themes through regular discussion with the wider research team, with areas of uncertainty resolved through consensus. No qualitative analysis software was used.

Patient and public involvement

Patient and public contributors, including nine women and birthing partners with lived experience of OEs, were recruited through Valuing Our Intellectual Capital and Experience (VOICE), an established public involvement network, using purposive sampling to maximise variation in experiences. They contributed to the development of the review question and inclusion criteria, advised on search scope and terminology and commented on emerging findings during synthesis. In addition, clinicians from trainee to consultant level contributed through a departmental learning event and a smaller group of clinicians provided ongoing input to sense-check the

developing findings throughout the review. This involvement aligns with the scoping methodology's intention to broaden the review by integrating the perspectives of those with knowledge of the area under examination.⁶ Further details are in online supplemental file S3.

RESULTS

Screening process

Searches returned 5021 papers initially and a further 925 when updated. Screening identified 263 papers initially and a further 106 when updated resulting in 369 sources included in the final review. The screening process is summarised in the PRISMA flowchart (online supplemental file S4).

Characteristics of included studies

All global regions were represented, reflecting the world-wide relevance of knowledge-building around OEs. Publication rates were relatively consistent across the review period. Studies were classified according to the primary geographical setting in which the research was conducted and the predominant study design or methodology reported by the authors (table 1). The geographical distribution highlights both the diversity of health system contexts in which OEs are managed and the potential for cross-context learning. A full summary of the characteristics of included studies and their thematic allocation is provided in online supplemental file S5. Detailed results are presented in tables 2 and 3 and online supplemental file S6A–D, which are cited in the relevant sections.

Overview of findings

Analysis generated three interconnected domains through which knowledge in OEs was acquired and applied: training, context and experience (figure 1). Across these domains, the literature identified both barriers and enablers to knowledge-building, although barriers were considerably more frequently reported than enablers. Training reflected structured and formal approaches to knowledge acquisition; context encompassed both local workplace realities and wider systemic influences; and experience captured knowledge developed through clinical encounters, reflection and insights gained relationally through patients and colleagues. Knowledge-building occurred across these domains through the interplay of knowledge 'received' from published sources, colleagues, educators and patients, and knowledge 'embodied' through personal clinical experience and practice (figure 1).

Knowledge within training

Formal training is the most consistently described source of knowledge for managing OEs. Simulation—delivered in both educational and workplace contexts—dominates the literature (online supplemental file S6A), encompassing methods from low-cost, improvised models (OE136⁸, OE130,⁹ OE215)¹⁰ to advanced, immersive

Table 1 Classification of included sources

	Original papers (OE1–263)	Updated search (OE264–369)
Classification based on 'continent of focus'		
Africa	65	23
Asia	70	32
Europe	57	24
International	25	16
Multi-continent	2	–
Oceania	11	–
North America	29	10
South America	4	1
Classification based on 'source type'		
Primary literature	n=228	n=89
Case study	28	11
Consensus	1	1
Descriptive	5	3
Evaluation	29	12
Experimental	7	5
Feasibility study	2	1
Implementation study	2	1
Methodological research	2	1
Mixed methods	8	4
Observational	70	20
Participatory	1	–
Pilot	2	–
Qualitative	28	8
Quality improvement	3	2
Quasi-experimental	9	3
Questionnaire	22	12
Secondary empirical data	9	5
Review literature	n=35	n=17
Document analysis	1	1
Narrative	14	9
Scoping	–	1
Systematic	19	6
Other evidence synthesis	1	–

virtual reality systems (OE104,¹¹ OE329),¹² with standardised commercial models being most common. Simulation can improve team performance (OE55)¹³ and show potential to contribute to better clinical outcomes (OE4).¹⁴ However, evidence for sustained behaviour change and clear demonstration of clinical impact remains limited (OE207,¹⁵ OE28,¹⁶ online supplemental file S6B). Workplace-integrated simulation offers additional advantages: minimising service disruption (OE230),¹⁷ enabling more frequent training (OE92),¹⁸ and creating authentic interprofessional learning (OE58,¹⁹ OE196,²⁰ OE119).²¹

Accounts of workplace-based learning (WPBL) from real-patient encounters—whether as part of formal training or arising organically from clinical work—are sparse. Blended approaches combining simulation

Table 2 Factors influencing facility preparedness impacting obstetric emergency (OE) care delivery

Domain	Description	References	Total
Resources	General preparedness including infrastructure	OE65, OE20, OE220, OE138, OE129, OE231, OE46, OE45, OE234, OE35, OE197, OE270, OE272, OE293, OE295, OE298, OE314, OE317, OE343, OE352, OE360, OE362, OE277	23
Drugs	Availability and quality	OE100, OE31, OE176, OE168, OE196, OE46, OE277	7
Staff	Inadequate numbers, high-turnover, absenteeism, unskilled, maldistributed	OE117, OE100, OE146, OE54, OE99, OE230, OE126, OE168, OE129, OE74, OE196, OE8, OE231, OE168, OE76, OE265, OE266, OE277, OE280, OE293, OE298, OE305, OE314, OE317, OE319, OE322, OE330, OE341, OE343, OE363, OE277, OE351	32
Blood products	Availability	OE117, OE196, OE176, OE322	4
Electricity	Reliability and backup generators	OE100, OE46	2
Water	Availability, reliability	OE46	1
Equipment	Availability, reliability	OE100, OE146, OE74, OE196, OE46, OE295, OE306, OE314, OE333	9
Supply chain	Unreliability	OE74, OE46	2
ICU beds	Availability	OE123	1
Printed guidelines/ protocols	Availability	OE277	1

ICU, intensive care unit; OE, obstetric emergency.

with real-world exposure are recognised as valuable (OE248),²² especially for rarer scenarios (OE199).²³ Few studies describe strategies to make this learning visible (OE308),²⁴ with assessment of knowledge being the primary focus (OE34,²⁵ OE178,²⁶ OE97,²⁷ OE227,²⁸ OE59,²⁹ OE7,³⁰ OE286,³¹ OE325).³² Similarly, while course evaluations are common—using tests (OE171,³³ OE195,³⁴ OE103,³⁵ OE159)³⁶ and novel self-assessment tools (OE226)³⁷ or video observation (OE149)³⁸—how such assessments themselves act as a source of knowledge, or inform future workplace practice, is under-investigated. Training on personal and professional responses to adverse OE outcomes is also rare, despite clear evidence of their emotional and psychological impact on providers (OE10,³⁹ OE13,⁴⁰ OE93).⁴¹

Knowledge acquisition from training is undermined when courses are poorly designed (OE195,⁴² OE177,⁴³ OE108),⁴⁴ misaligned with policy (OE144),⁴⁵ unaffordable (OE162,⁴⁶ OE2),⁴⁷ or inaccessible (OE32).⁴⁸ Even high-quality training may fail to translate into practice when personal factors intervene; resistance to change (OE95),⁴⁹ fear of losing autonomy (OE134),⁵⁰ or conflicts between personal beliefs and best practice (OE99).⁵¹ Decision-making is also shaped by experience, individual rationale (OE208)⁵² and some personality traits—extraversion and neuroticism—particularly pronounced among O&G doctors (OE67),⁵³ which may contribute to variation in OE outcomes. Without adequate resources, birth in a healthcare facility offers no survival advantage (OE206,⁵⁴ OE197).⁵⁵ If training does not account for local human (OE262)⁵⁶ or material (OE45)⁵⁷ resource constraints, staff are forced to improvise (OE231,⁵⁸ OE45⁵⁷), limiting the application of knowledge.

Organisational barriers exist (table 2)—including absent systems, guidelines and governance (OE100⁵⁹,

OE32),⁶⁰ entrenched hierarchies (OE231⁵⁸), poor patient rapport (OE96⁶¹) and, in some contexts, corruption, violence and cultural norms (OE138)⁶²—that further impede implementation. Examples include punitive cultures (OE231⁵⁸) including staff beatings after adverse outcomes resulting in fear of involvement (OE138⁶²); corruption through direct payments for deliveries, unnecessary interventions or sale of equipment (OE138⁶²); and cultural norms shaping care, such as preference for male babies resulting in reluctance to intervene for female fetuses (OE138⁶²), avoidance of sterilisation postuterine rupture repair to preserve marital relations despite health risks (OE99⁵¹) and restrictions on male doctors attending births (OE138⁶²).

Conversely, training that is context-specific, culturally sensitive and tailored to available resources (OE89,⁶³ OE138⁶², OE248⁶⁴) is more likely to be applied in practice. High participation rates enhance short-term impact (OE187,⁶⁵ OE95,⁶⁶ OE94),⁶⁷ while sustained benefits are supported by supervision (OE262,⁵⁶ OE76,⁶⁸ OE230),⁶⁹ mentorship (OE58,¹⁹ OE150,⁷⁰ OE138⁶²) and exposure to varied settings (OE252⁷¹). Supervision is particularly critical, with evidence of increased complications during periods of reduced oversight—such as nights, weekends, and staff changeovers (OE20,⁷² OE237,⁷³ OE217,⁷⁴ OE220).⁷⁵ Finally, safe workplace learning cultures (OE231⁵⁸) both enable the application of knowledge acquired through formal training and provide fertile ground for further development through reflective, real-time practice.

Knowledge within context

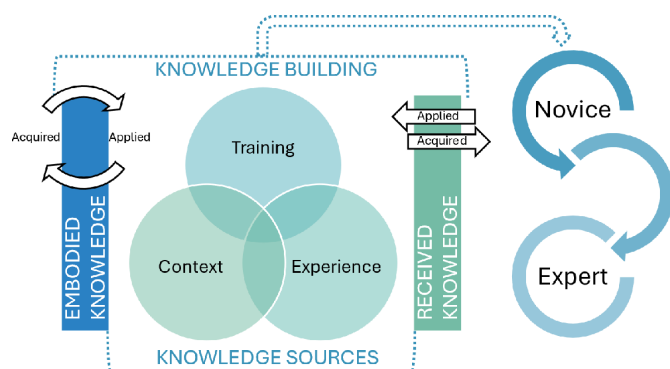
Knowledge within context emerges in two main forms: awareness of the local environment in which care is

Table 3 Factors influencing delays in accessing care for obstetric emergencies (OEs), categorised as barriers and enablers

Barriers		Enablers	
Patient perceptions and decision-making			
Poor knowledge of danger signs	OE90, OE111, OE52, OE14, OE129, OE265, OE266	Community education programmes	OE57, OE243, OE213, OE220, OE10, OE105, OE115, OE266
Uncertainty about when to access care	OE105, OE166, OE220.		
Complication readiness	OE118, OE111, OE52, OE121. OE265, OE266		
Cost	OE71, OE54, OE209, OE249.		
Negative facility experience	OE175, OE213.		
Cultural beliefs	OE213, OE57, OE196, OE129.		
Accessibility of facilities			
Facilities unavailable	OE51, OE220, OE129, OE176, OE72, OE121, OE33, OE209, OE352.	Expansion of ambulance services	OE213
Lack of transport	OE105, OE65, OE72, OE11. OE76, OE278, OE293	Accurate travel times	OE29, OE151, OE256, OE360
Vast geography	OE72, OE352, OE360	Maternity wait homes	OE249, OE65, OE255
Poor road	OE213		
Mobile reception to access transport	OE213		
Triage and referral effectiveness			
Failure to recognise deterioration	OE166, OE134, OE237, OE322, OE330, OE341, OE277	Standardised triage proformas	OE62, OE133, OE247. OE219, OE298
Poor infrastructure	OE202, OE314	Maternity specific observation parameters	OE212, OE244, OE250
Facility preparedness	OE73, OE220, OE11, OE314, OE317, OE277	Integrated and standardised escalation triggers	OE219. OE41, OE165, OE343
Poor communication	OE146, OE129, OE241, OE220, OE237, OE20, OE213, OE217, OE51, OE213, OE54, OE121, OE259, OE176, OE179, OE270, OE293, OE314		
Unclear guidelines/processes	OE129, OE241, OE220, OE237, OE20, OE213, OE217, OE51, OE54, OE121, OE11, OE270, OE293		

delivered, and engagement with the wider professional and academic landscape.

Locally, understanding the specific constraints and enablers of a setting allows O&G doctors to anticipate


Figure 1 Interconnected sources of knowledge and processes of knowledge-building around obstetric emergencies.

and address barriers to care. Measuring preparedness (OE26⁷⁶, OE21,⁷⁷ OE6)⁷⁸ and recognising the critical points where delays occur—recognition of an emergency by the patient and decision to seek care, access to facilities and timely recognition and escalation by healthcare providers (OE114,⁷⁹ table 3)—can inform targeted strategies. Building this knowledge enables clinicians to prepare women during consultations (OE65),⁸⁰ advocate for more equitable facilities (OE147,⁸¹ OE257)⁸² and influence policy change. Codesigned educational strategies within communities can help tailor care, improve access and build relationships (OE57,⁸³ OE243,⁸⁴ OE255).⁸⁵

Awareness of the wider context is constructed largely through engagement with literature, offering both practice and outcome-focused insights. Practice-focused sources (online supplemental file S6C)—including empirical studies, case reports, and literature reviews—introduce new technical and, to a lesser extent, non-technical skills, test the feasibility of interventions and

share innovations in monitoring, imaging and acute management and, increasingly, the application of artificial intelligence in obstetric care (OE274,⁸⁶ OE289).⁸⁷ Process evaluations, such as those measuring decision-to-delivery interval, provide a further lens on OE management, revealing both personal and systemic influences.

Outcome-focused literature (online supplemental file S6D) examines incidence and associated factors drawing attention to inequalities (OE43,⁸⁸ OE176,⁸⁹ OE52,⁹⁰ OE121⁹⁰) and modifiable contributors to poor care. Insights come from both local reviews and large-scale national surveillance programmes, which can identify practice variation (OE110)⁹¹ and enable learning from rarer events where clinical trials are impractical (OE213,⁹² OE110⁹¹).

Together, these sources refine risk assessment, inform preventive strategies and target interventions towards vulnerable groups. They contribute to a deeper understanding of the conditions in which OEs occur and the strategies most likely to improve them, helping O&G doctors shape both immediate decisions and long-term professional growth.

Despite the potential value of context in knowledge-building, gaps remain in understanding how, or if, O&G doctors' access and use these sources. Engagement with literature is often hindered by paywalls, ability or competing demands. Locally, constraints on resources, staffing and equipment create barriers to the application of newly acquired knowledge, even when O&G doctors are aware of best practice.

Knowledge within experience

Knowledge within experience develops through clinicians' reflection on OEs and through insights gained from patient and colleagues. Experiential knowledge may arise from direct clinical encounters, shared stories or structured reflections. Debriefing, in particular, is widely used in simulation training and increasingly valued after real events for its ability to promote shared learning, provide emotional support and improve future care (OE228⁹³).

Women's experiences provide essential insights for improving care (OE350).⁹⁴ OEs are associated with high rates of trauma among women (OE148,⁹⁵ OE196²⁰, OE93⁴¹). While many women perceive facilities as well equipped (OE210⁹⁶), concerns about understaffing, inadequate supervision and the risk of mistreatment (OE175,⁹⁷ OE213⁹²) highlight systemic weaknesses that undermine safety and trust.

Consent is one area where O&G doctors' and women's perspectives may diverge (OE131⁹⁸, OE225⁹⁸, OE172).⁹⁹ O&G doctors often describe written consent as a medicolegal safeguard (OE98),¹⁰⁰ despite sometimes questioning a woman's capacity to consent during an OE (OE216,¹⁰¹ OE135).¹⁰² In contrast, many women report trusting clinicians to act in their best interests (OE196²⁰, OE53¹⁰³) valuing empathetic, personalised communication over formal processes (OE66).¹⁰⁴ Attention to

these perspectives may help shape more responsive, patient-centred approaches to consent in OEs (OE131⁹⁸, OE279¹⁰⁵).

Despite its importance, experiential knowledge is underrepresented in the literature. Few studies capture the personal experiences of O&G doctors and patient narratives are similarly limited. This restricts opportunities to learn from lived experience and refine practice through reflection on real events.

Opportunities for experiential learning are also shaped by workplace conditions. Barriers to debriefing—including high workload, limited space, a shortage of skilled facilitators and the unpredictability of OEs—limit opportunities to share learning from experience (OE196²⁰). Blame cultures further hinder open discussion, intensifying emotional burdens (OE10³⁹), discouraging honest reflection and driving defensive practice (OE93⁴¹, OE19¹⁰⁶, OE263.¹⁰⁷

Conversely, structured, blame-free forums for reflection can support learning from experience. Systems that prioritise learning over punishment enable more open discussion of adverse outcome (OE86).¹⁰⁸ For example, Taiwan's no-fault compensation scheme—which assigns monetary values to adverse birth outcomes—encourages open discussion and reduces fear of personal repercussions (OE86¹⁰⁸). In Denmark, video-assisted debriefing after real-life OEs was initially met with concerns about exposing weaknesses but was ultimately valued for its educational benefits (OE128).¹⁰⁹ When embedded within a safe learning culture, these approaches can transform individual experiences into collective knowledge that strengthens OE preparedness.

DISCUSSION

Main findings

This review aimed to identify how O&G doctors build knowledge in OEs and how these insights could inform UK-based improvement. It addressed two research questions: (1) what sources can O&G doctors' use to build knowledge around OEs? and (2) what barriers exist to building and applying knowledge?

Knowledge-building sources clustered into training, context and experiences, with formal training remaining the primary focus of the literature. Barriers to learning were both structural, such as access to training, knowledge and resources, and cultural, including blame, punitive attitudes and entrenched hierarchies. Inequalities permeated all domains, driving inequities in outcomes. Limited access to care, transport, facilities and continuing education, coupled with poor staffing and resourcing, created cycles where the most disadvantaged communities received the poorest care.

Interpretations

In this review, knowledge-building is understood as the processes through which O&G doctors acquire, develop

and apply knowledge relevant to OEs across training, context and experience. Our analysis reveals constraints in current practice and identifies two key opportunities for improvement: expanding learning beyond formal training and developing collective knowledge through shared experience.

Reimagining learning

The findings of this review suggest that knowledge-building in OEs extends beyond the acquisition of technical skills through simulation. While simulation has an established role^{110–112} and increasingly incorporates workplace, longitudinal and sociocultural approaches to learning, it should coexist with—and be complemented by—ongoing WPBL, which remains under-represented in the literature despite its central role in knowledge-building.^{113–115}

Emerging technologies, including artificial intelligence, are increasingly used to enhance simulation-based training.^{116–118} While these approaches may offer benefits, an exclusive focus on technologically mediated learning risks overlooking the educational value of real clinical environments.^{114 115 119} These findings suggest value in considering how WPBL and simulation may complement one another, although further research is needed to understand how best to integrate these approaches.^{113 116 118} WPBL can take two forms: in situ simulation and real patient encounters. In situ training, delivered with the actual teams and in the real clinical environment, mirrors the final ‘performance’ and may improve authenticity and team preparedness.^{14 120} Learning with real patients—long the foundation of medical training—is less openly discussed, particularly in maternity care, where concerns about consent, privacy and patients feeling they are being used primarily for training can make its educational role difficult to acknowledge openly.¹²¹ This taboo limits its potential. The limited literature describing learning from real patient encounters suggests this may represent an underdeveloped area of OE education. Future work could explore how feedback, rehearsal and reflection might be more explicitly incorporated into everyday clinical practice alongside existing simulation-based approaches. Training programmes should treat WPBL as an active process, integrating rehearsal, feedback and reflection into everyday practice rather than confining it to isolated sessions or technology-centred simulation.^{113–115 119}

The reviewed literature also highlighted inequalities in OE outcomes and access to training. These findings demonstrate the importance of considering whether educational content sufficiently reflects the diversity of populations served, particularly within UK maternity care. In the UK, medical resources frequently under-represent conditions in non-white skin,¹²² and simulated scenarios often default to white patients, overlooking cultural and clinical contexts relevant to racially minoritised groups. Although the included studies primarily described inequalities in access, outcomes and resources rather than examining specific structural determinants, broader consideration of social and structural influences on OE

care, including racism, poverty, ableism and cisnormativity, may strengthen educational approaches within the UK context.^{123 124} Embedding these perspectives into OE education, particularly in debriefs led by skilled, sensitive facilitators, could improve cultural competence and reduce inequities.

Our synthesis suggests that contextual knowledge may support clinicians in identifying opportunities for service improvement and addressing local barriers to OE care. Such engagement enables them to advocate for system improvements, address inequities and strengthen OE preparedness. Barriers to care create barriers to learning—when women cannot access timely services, opportunities for clinicians to learn from real-world OEs are lost, and with them, opportunities to save lives.

Taken together, these findings suggest several opportunities for future development. These include further exploration of WPBL alongside simulation, greater attention to contextual and equity-related factors within educational design, and evaluation of educational outcomes beyond technical performance alone (OE4¹⁴). WPBL has long been shaped by a perceived tension between service and training, yet positioning these domains as competing priorities may risk obscuring their mutual dependence and diminishing the educational value inherent in everyday clinical work.^{114 115 119} Analysis of the NHS Long Term Workforce Plan¹²⁵ concluded education must stand alongside service and research in practice and policy to support both world-class training and safe, effective patient care.¹²⁶ This requires integrated models of clinical work recognising service delivery itself as a site of learning and knowledge production.^{114 115} Recent policy developments¹²⁷ begin to acknowledge this shift by advocating for clinical education to be normalised as a central, resourced, timetabled and appraised component of professional practice. However, the reviewed literature suggests learning in OEs continues to be commonly conceptualised as episodic, emerging primarily through isolated training interventions or postevent reflection following adverse outcomes, rather than as an ongoing feature of everyday clinical work. This reinforces a focus on ‘objective training’ and external performance expectations, while overlooking the relational, contextual and uncertain nature of knowledge construction within complex workplaces.^{128–131} Situating learning in the broader ecology of the workplace positions everyday practice as a powerful site for knowledge-building—laying the groundwork for the shared, cross-boundary learning described below.

Shared experience, shared understanding

The findings also suggest opportunities to strengthen knowledge-building through greater integration of patient and multidisciplinary perspectives. Contrasting perceptions of childbirth between women, O&G doctors and midwives¹³² reveal misalignment but collaborative learning spaces—rarely prioritised—could foster a shared understanding that strengthens practice.

Women report feeling poorly informed about the rationale for emergency interventions^{133 134} and experience high levels of psychological trauma^{135 136} leading to long-term sequelae including breast feeding difficulties and worsening mental health.¹³⁷ The limited literature describing women's experiences, together with the high prevalence of birth-related trauma identified in this review, suggests two areas warranting further exploration: improving antenatal preparation and understanding how patient narratives may contribute to clinician learning. Patient narratives may offer additional insights into cultural norms, barriers to help-seeking and perceptions of positive birth experiences that are less visible within clinician-focused literature.¹³⁸ Safety must be defined beyond mortality, challenging abusive or non-dignified practices disproportionately affecting racially marginalised and socioeconomically disadvantaged groups.^{139 140} Despite well-documented institutional racism and poorer outcomes in UK maternity care,¹⁴¹ racialised perspectives remain rare in UK OE literature, though those that exist from other settings offer valuable insights.¹⁴² Codesigned

educational approaches involving patients and families warrant further exploration and may offer multiple benefits.^{143 144}

OEs negatively affect staff well-being,¹⁴⁵ yet personal experiences—particularly of O&G doctors—are rarely captured in academic literature, perhaps reflecting discomfort with exposing systemic failings. The popularity of memoirs detailing an O&G doctors' OE experiences¹⁴⁶ suggests public interest, despite a polarised reception.¹⁴⁷ Informal storytelling has long supported learning,¹⁴⁸ but fear of litigation continues to constrain honest disclosure. Midwives' accounts reveal toxic cultures, institutional disrespect and cycles of blame¹⁴⁹—in extremes mirroring the violence experienced by patients⁶²—evidencing the manifestation of blame in psychological and physical sequelae. Collectively, these findings suggest that psychologically safe, reflective workplace cultures may support both staff well-being and organisational learning.

Although most OE workplace learning literature originates from LMICs, clear parallels exist with the UK. Staff improvisation under resource constraints⁵⁸

Table 4 Recommendations to optimise knowledge-building in obstetric emergencies in the UK

	Recommendation	Rationale
Policy	Explore approaches to operationalising WPBL within national OE training programmes by formalising educational strategies, creating content to explicitly 'train the trainer' and embedding learning visibly into real clinical settings alongside simulation.	The reviewed literature suggests WPBL may support context-sensitive knowledge-building and team readiness. However, it remains poorly defined and largely assumed to occur without explicit direction in UK policy. Structured implementation could enhance its reach and impact.
	Consider embedding equity considerations into all OE training by integrating cultural competence, escalation bias awareness, trauma-informed care, psychological safety and patient-centred communication.	Helpful in addressing systemic inequities, reflects diverse patient needs and cultural norms, and tackles widespread gaps in communication and trauma prevention in current OE education.
	Take steps towards establishing nationally coordinated, co-produced, workplace focused, OE educational strategies, bringing interdisciplinary staff groups and patients together to design and deliver shared curricula, drills and scenario planning.	Supports the inclusion of all voices in shaping training, aligns care with patient-defined positive birth experiences, aligns expectations with mutual understanding, reduces blame, and strengthens empathy between staff and patients.
Practice	Consider ways in which simulation provision could shift to emphasise in-situ drills conducted in authentic teams, settings and conditions.	Real-world rehearsal has been associated with improved skill transfer, teamwork and patient safety.
	Embed teaching on social and systemic determinants of health—such as racism, poverty, ableism and cisnormativity—into all OE education, ensuring delivery and debriefs are led by skilled, sensitive facilitators and grounded in local context.	Understanding the cultural and systemic realities of the population served enables clinicians to tailor care, improve cultural competence and address inequities in OE outcomes.
	Support structured reflective spaces after OEs—such as debriefs or Schwartz rounds—to support facilitate experiential learning and emotional processing.	Facilitated reflection can promote staff well-being, encourages a growth mindset and transforms lived experiences into improved preparedness for future emergencies.
	Improve equitable access to OE training by providing protected time, dedicated funding, and adopting low-cost, in-situ models proven effective in lower-resourced settings.	Ensuring training reaches rural, under-resourced, or high-turnover units reduces preparedness gaps, while low-cost, context-specific approaches maximise impact and make funding go further.
Research	Evaluate the impact of educational strategies that support WPBL with real patients, incorporating a strong focus on context to understand what works, for whom, and in which settings.	There is limited evidence on the real-world effectiveness of OE education beyond simulation. Understanding how WPBL performs across diverse contexts (such as rural vs urban units, high-resource vs low-resource settings or varying team compositions) -will ensure strategies are tailored, scalable and relevant to local needs.
	Investigate racial and social inequalities in OE care experience and outcomes.	Marginalised voices remain scarce in the literature despite poorer outcomes and documented institutional bias.
	Study how knowledge flows between professions and from patients to clinicians in OE settings.	Understanding these pathways will help assist in designing cohesive, inclusive learning systems.

OE, obstetric emergency; WPBL, workplace-based learning.

echoes UK experience, and lessons from settings that have thrived despite fewer resources—including in situ training models¹⁵⁰—are highly relevant given current workforce shortages. International comparisons also highlight the value of national surveillance systems, such as MBRRACE-UK, for promoting mutual knowledge exchange.¹⁵¹ Holistic education must bridge countries, facilities and institutions, recognising that even small steps towards shared learning can generate meaningful local improvements. Greater integration of learning across professional groups, settings and patient perspectives may strengthen OE preparedness. Patients and staff experiences remain under-represented in the literature, yet their inclusion could help shape more practical and context-sensitive educational approaches.

CONCLUSION

UK maternity care faces mounting pressure to improve outcomes and experiences.^{2 3 5} O&G doctors draw on multiple sources to build knowledge around OEs, yet WPBL and knowledge generated through clinical experience remain under-represented in research and training. Learning is commonly framed as the transfer of knowledge through discrete training events, rather than as an ongoing process embedded within everyday clinical work. The limited representation of WPBL within the literature suggests opportunities to better understand how learning embedded within everyday clinical practice contributes to OE knowledge-building. Developing and rigorously evaluating WPBL approaches, therefore, represents an important area for future research. Developing and rigorously evaluating WPBL approaches in OE is, therefore, an important next step. Strengthening OE education will require approaches that are sensitive to context, address inequities in access to training and care and support learning across professional groups and settings.

Recommendations for future policy, practice and research

Although this review synthesised international evidence, the aim was not to identify context-specific solutions but to map how doctors build-knowledge from and through OEs across diverse settings. Many of the identified themes were consistently reported across healthcare systems and resonated with our stakeholder discussions involving UK clinicians and people with lived experience, supporting their relevance to UK maternity education and workforce strategies. However, implementation should always be tailored to local organisational and healthcare contexts.

Drawing on the breadth and consistency of themes identified across the mapped literature, table 4 summarises priorities that emerged from this review for future policy, practice and research. Consistent with scoping review methodology, no formal quality appraisal was undertaken, so recommendations reflect the breadth and consistency of the available evidence rather than its strength or certainty. Given this, they should be interpreted as areas warranting further consideration.

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